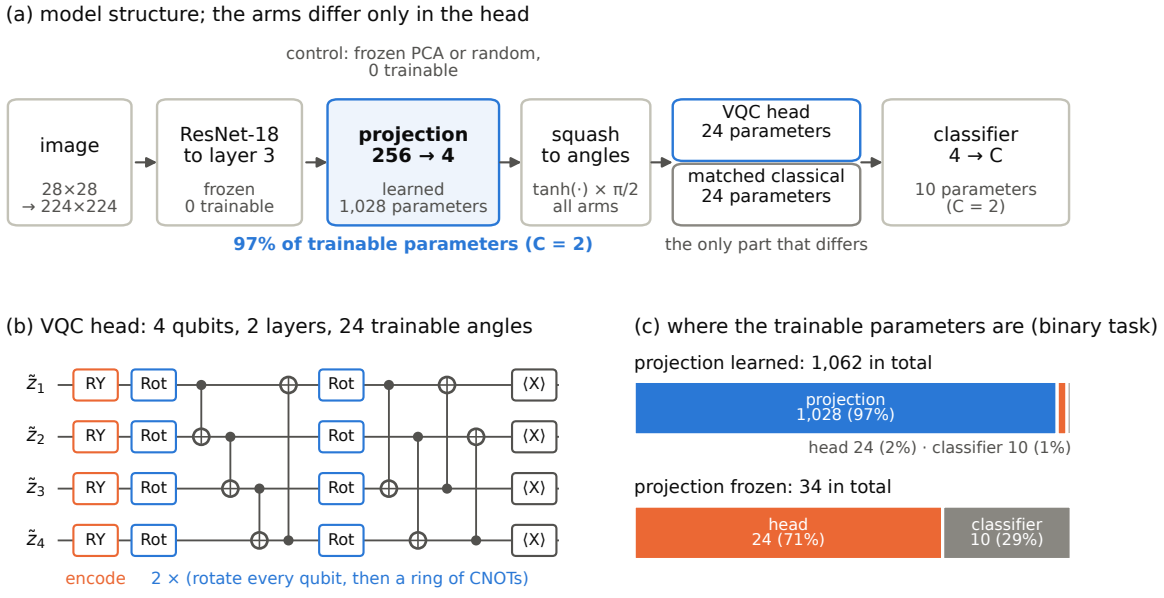
(a) model structure; the arms differ only in the head
control: frozen PCA or random, 0 trainable
image
28×28 → 224×224
ResNet-18 to layer 3
frozen 0 trainable
projection 256 → 4
learned 1,028 parameters
squash to angles
tanh(·) × π/2 all arms
VQC head 24 parameters
matched classical 24 parameters
classifier 4 → C
10 parameters (C = 2)
97% of trainable parameters (C = 2)
the only part that differs
(b) VQC head: 4 qubits, 2 layers, 24 trainable angles
RY
Rot
⟨X⟩
encode
2 × (rotate every qubit, then a ring of CNOTs)
(c) where the trainable parameters are (binary task)
projection learned: 1,062 in total
projection 1,028 (97%)
head 24 (2%) · classifier 10 (1%)
projection frozen: 34 in total
head 24 (71%)
classifier 10 (29%)